

SURVIVAL DAYS: A ZOMBIE SURVIVAL SIMULATOR



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"EVERY SURVIVOR NEEDS A PLAN"

ABOUT THE PROJECT

MY APPROACH

- Build Survival Days around three classes that must cooperate: Player, Zombie, and Game.
- Keep it small enough to finish in a course timeline, but structured enough to genuinely exercise encapsulation, composition, and constructors.
- Wrap it in a zombie-survival theme so the mechanics stay motivating to build and to play.



"KNOW YOUR ENEMY BEFORE THE BATTLE"

PROGRAM OVERVIEW

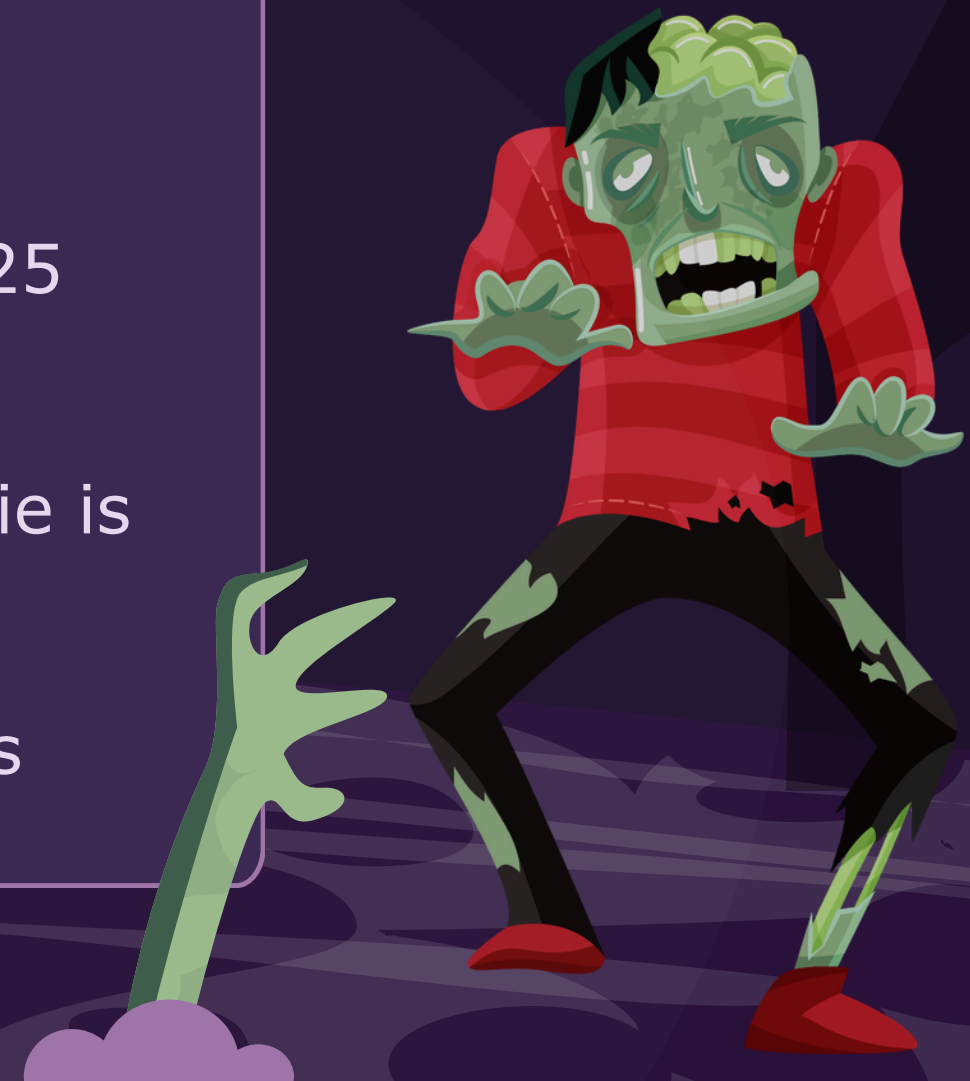
OOP Concept	What It Means in Survival Days
Class	Player, Zombie, and Game — the three cooperating actors
Encapsulation	Health, hunger, food, bullets are private; only reachable via methods
Composition	Game owns one Player and creates Zombie objects on demand
Constructor	Each class sets its own starting state the moment it's created
Method	eat(), shoot(), attack(), damageTaken() — controlled behavior



"MY WEAPONS FOR THE APOCALYPSE"

FEATURES

- Four interdependent resources (health, hunger, food, bullets) tracked every single day
- A turn-based loop: search, fight a zombie, eat, or quit
- Random search outcomes: food, bullets, nothing, or a surprise zombie ambush
- Every zombie spawns with a randomized attack strength (10–25 damage)
- A combat loop that resolves shot-for-shot until player or zombie is defeated
- Automatic game-over detection when health or hunger reaches zero, reporting days survived



"THE CAST OF SURVIVORS"

CLASSES & METHODS

Player

ATTRIBUTES

name, health,
hunger, bullets, food

METHODS

- eat()
- damage(amt)
- loseHunger()
- shoot()
- addFood()/addBullets()
- showStatus(day)

Zombie

ATTRIBUTES

health, damage

METHODS

- damageTaken()
- attack(Player&)
- getHealth()

Game

ATTRIBUTES

Player player,
int day

METHODS

- start()
- search()
- fight()
- fightZombie(Zombie&)

Game owns one *Player* for the session and creates a new *Zombie* every time a fight breaks out — that has-a / creates relationship is composition.



"SKILLS I UNLOCKED IN C++ PROGRAMMING"

KEY CONCEPTS USED

- **Encapsulation** — private state accessed only through public methods like eat() and shoot()
- **Composition** — Game has-a Player and creates Zombie objects as encounters occur
- **Constructors** — each class initializes its own starting values on creation
- **Reference parameters** — attack(Player &player) lets Zombie modify the real Player object
- **Nested control flow** — while loops inside start() and fightZombie() drive turns and combat
- **rand() / srand()** — seeded randomness for search outcomes and zombie damage range

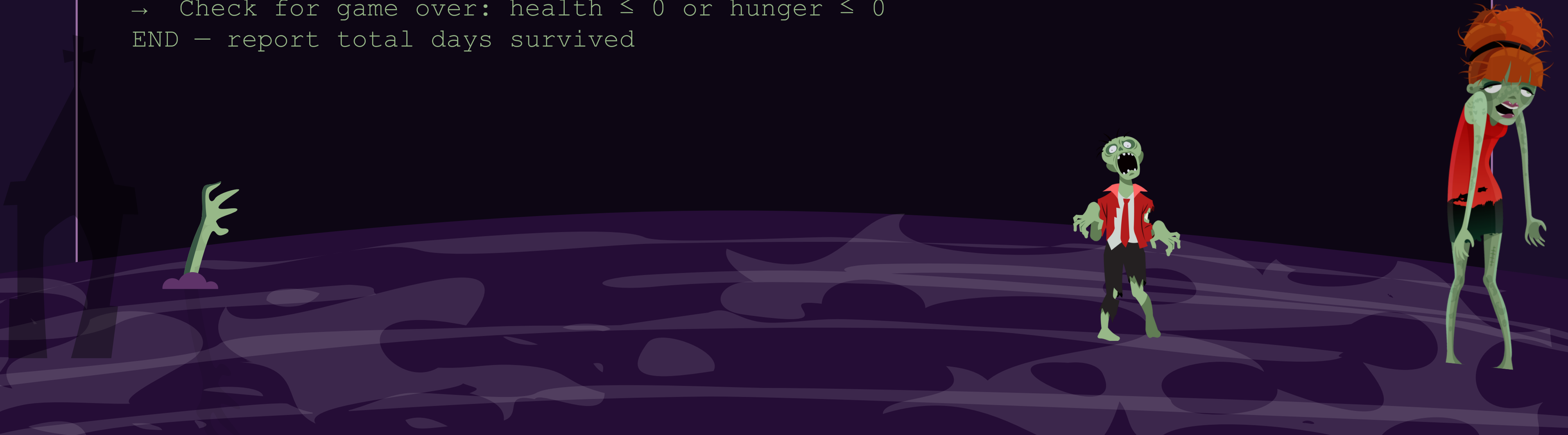


"MY HERO'S JOURNEY THROUGH THE CODE"

PROGRAM FLOW

START

- Ask for the player's name and create the Player and Game objects
 - Loop: show status, then choose Search / Fight / Eat / Quit
 - Resolve the chosen action, updating health, hunger, food, or bullets
 - Apply passive hunger decay and advance the day counter
 - Check for game over: $\text{health} \leq 0$ or $\text{hunger} \leq 0$
- END – report total days survived



"THE BATTLE REPORT HAS ARRIVED"

SAMPLE OUTPUT

--- Day 3 ---

Health: 70

Hunger: 60

Food: 1

Bullets: 3

1. Search 2. Fight Zombie 3. Eat Food 4. Quit

Choose: 1

Oh no! A zombie found you!

You shot the zombie!

You defeated the zombie!

You found 1 food!



"NO TWO NIGHTS ARE THE SAME"

RANDOMIZATION & REPLAYABILITY

Search Outcomes

Each search rolls 1 of 4 equally-likely results: food, +2 bullets, nothing, or a zombie ambush.

Zombie Strength

$\text{rand()} \% 16 + 10$ gives every zombie a fresh attack value between 10 and 25 damage.

Why It Matters

No two playthroughs feel the same, so resource planning stays meaningful run after run.



"QUEST COMPLETE: WHAT I ACHIEVED"

CONCLUSION

- A fully playable, turn-based C++ console game built from scratch
- Three cooperating classes — Player, Zombie, Game — demonstrating encapsulation, composition, and constructors
- Randomized events and combat give the game genuine replay value
- Reinforced control flow, state management, and defensive programming skills



"The Graveyard Bows Before You..."

THANK YOU!

Questions about Survival Days?

